

Homogeneous Systems

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Every homogeneous system has a **trivial solution** where $\mathbf{x} = \mathbf{0}$.

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Theorem

The homogeneous equation $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution if and only if the solution set has at least one free variable.

Solution Sets of Consistent Systems

Find the solution set to the homogeneous system write in vector form (from lecture 4):

$$x_1 + 3x_2 - 2x_3 = 0$$

$$-2x_1 - 6x_2 + 4x_3 = 0$$

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More Practice!

Find the solution set to the **nonhomogeneous** system

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Which can be written as

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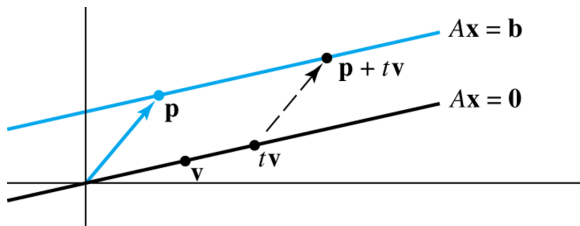
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Infinite Solutions from Homogeneous System

Theorem

Let $A\mathbf{x} = \mathbf{b}$ be a consistent system with solution $\mathbf{x} = \mathbf{p}$. Then there exists a solution set of vectors of the form

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Where $t\mathbf{v}$ is a solution to the *homogeneous* system $A\mathbf{x} = \mathbf{0}$.

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